

Analysis 1: Loyalty Segment Distribution

Q.What type of customers does the company currently have?

SQL Query:-

```
SELECT loyalty_segment,  
COUNT(*) AS total_customers  
FROM customer_intelligence_dataset  
GROUP BY loyalty_segment;
```

Output:-

loyalty_segment	total_customers
Loyal	709
Potential Loyalist	716
Champion	687
At Risk	657

Insight:

Potential Loyalists represent the largest customer segment (716 customers), followed closely by Loyal customers (709). This indicates a strong opportunity to convert existing customers into high-value Champion customers through targeted retention strategies

Recommendation:

Prioritize conversion campaigns for Potential Loyalists, as they represent the largest segment and have the highest potential to become Champions. Simultaneously, maintain engagement programs for existing Loyal and Champion customers to prevent churn

Analysis 2: Customer Pyramid

Q. How is the customer base distributed?

SQL Query:-

```
SELECT value_tier,  
COUNT(*) AS total_customers  
FROM customer_intelligence_dataset  
GROUP BY value_tier;
```

OUTPUT:-

value_tier	total_customers
Very Low Value	677
Medium Value	696
High Value	691
Low Value	705

Insight:

The customer base is distributed almost evenly across all value tiers, with no single segment dominating the population. This indicates diverse spending behaviors and customer values, creating opportunities for targeted strategies rather than a one-size-fits-all approach

Recommendation:-

Implement tier-specific engagement strategies:-

- **High Value customers:** Offer exclusive rewards and personalized recommendations.
- **Medium Value customers:** Encourage upselling
- **Low Value customers:** Focus on increasing purchase frequency through targeted campaigns

Analysis 3: Loyal vs Discount Dependent Customers

Q. Who are genuinely loyal customers and who only buy with discounts?

SQL Query:-

```
SELECT dependency_segment,  
loyalty_segment,  
COUNT(*) AS customers  
FROM customer_intelligence_dataset  
GROUP BY dependency_segment, loyalty_segment  
ORDER BY customers DESC;
```

OUTPUT:-

dependency_segment	loyalty_segment	customers
Highly Promo Dependent	Champion	539
Loyal Customer	At Risk	519
Loyal Customer	Potential Loyalist	517
Loyal Customer	Loyal	392
Highly Promo Dependent	Loyal	317
Highly Promo Dependent	Potential Loyalist	199
Loyal Customer	Champion	148
Highly Promo Dependent	At Risk	138

Insight:

The analysis reveals two critical findings. First:- Large number of Champion customers (539) remain highly dependent on promotions, suggesting retention may be supported by discounts rather than pure brand loyalty.

Second:-A substantial number of customers classified as loyal are simultaneously categorized as At Risk (519), indicating that previous loyalty does not guarantee future retention.

Recommendation:-

1. Gradually reduce heavy discounting for Champion customers while replacing discounts with loyalty rewards and exclusive benefits.
2. Prioritize re-engagement campaigns for Loyal but at risk customers through personalized communication, targeted offers, and retention programs before they become inactive.

Analysis 4: High Value vs Low Value Customers

Q. What separates high-value customers from low-value ones?

SQL Query:-

```
SELECT value_tier,  
AVG(frequency_score) AS avg_frequency,  
AVG(loyalty_score) AS avg_loyalty,  
AVG(customer_value_score) AS avg_value  
FROM customer_intelligence_dataset  
GROUP BY value_tier;
```

OUTPUT:-

value_tier	avg_frequency	avg_loyalty	avg_value
Very Low Value	6.011816839	0.304664486	0.22601363
Medium Value	5.942528736	0.47169707	0.56392377
High Value	5.962373372	0.584632779	0.76842309
Low Value	5.835460993	0.410156788	0.4267792

Insight:-

Although purchase frequency remains relatively consistent across all value tiers, loyalty scores increase significantly from Very Low Value customers (0.305) to High Value customers (0.585). This suggests that customer loyalty is a stronger driver of customer value than purchase frequency alone. Customers who develop stronger brand loyalty tend to generate substantially higher overall value.

Recommendation:-

The company should prioritize loyalty-building initiatives rather than focusing solely on increasing purchase frequency. Personalized rewards, membership benefits, exclusive product access, and retention campaigns can help move Medium and Low Value customers toward the High Value segment.

Analysis 5: Best product categories

Q. Which categories drive stronger customer retention?

SQL Query:-

```
SELECT category,  
AVG(loyalty_score) AS avg_loyalty,  
AVG(customer_value_score) AS avg_value  
FROM customer_intelligence_dataset  
GROUP BY category  
ORDER BY avg_loyalty DESC;
```

OUTPUT:-

category	avg_loyalty	avg_value
Footwear	0.456294957	0.5086666
Accessories	0.4496111	0.50247608
Clothing	0.436530758	0.49618097
Outerwear	0.432299201	0.4648126

Insight:-

Footwear shows highest average loyalty score (0.456) and customer value score (0.509), making it the strongest-performing product category. Accessories also has relatively high loyalty and value, while Outerwear shows the lowest performance across both metrics. This suggests that certain product categories are more effective at fostering long-term customer relationships and generating customer value.

Recommendation:-

The company should prioritize investments and campaigns around the Footwear and Accessories categories, as these segments generate the highest loyalty and value. For Outerwear, further investigation is needed to understand whether product assortment, pricing, seasonality, or customer experience factors are limiting customer engagement.

Analysis 6: Geography Analysis

Q. Which locations show strongest organic demand?

SQL Query:-

```
SELECT location,  
AVG(dependency_score) AS avg_dependency,  
AVG(loyalty_score) AS avg_loyalty  
FROM customer_intelligence_dataset  
GROUP BY location  
ORDER BY avg_dependency ASC;
```

OUTPUT:-

Top States with Lowest Dependency Score (Organic Demand)

Location	Avg Dependency	Avg Loyalty
Kansas	0.273	0.398
Maine	0.321	0.420
Connecticut	0.327	0.400
Arizona	0.333	0.446
New Jersey	0.341	0.415

Top States with Highest Dependency Score (Discount Driven)

Location	Avg Dependency	Avg Loyalty
South Carolina	0.500	0.449
Oklahoma	0.509	0.422
Hawaii	0.522	0.465
Indiana	0.550	0.446
Iowa	0.558	0.490

Insight:

States such as Kansas, Maine, and Connecticut exhibit the lowest dependency scores, indicating stronger organic demand and a lower reliance on discounts to drive purchases. Conversely, Iowa, Indiana, Hawaii, Oklahoma, and South Carolina demonstrate considerably higher dependency scores, suggesting that customer purchases in these regions are more influenced by promotional offers.

Recommendation:-

Focus marketing investments on high-organic-demand states such as Kansas, Maine, and Connecticut, where customers purchase with minimal promotional influence. For highly discount-dependent regions such as Iowa, Indiana, and Oklahoma, gradually reduce discount dependency through loyalty programs, personalized offers, and value-based engagement strategies.

Analysis 7: Demographic Analysis

Q. Which age groups are most valuable?

SQL Query:-

```
SELECT age_group,  
AVG(customer_value_score) AS avg_value,  
AVG(loyalty_score) AS avg_loyalty  
FROM customer_intelligence_dataset  
GROUP BY age_group  
ORDER BY avg_value DESC;
```

OUTPUT:-

age_group	avg_value	avg_loyalty
18-25	0.50073896	0.447822058
46-55	0.50065953	0.448247789
26-35	0.49835489	0.440600394
55+	0.49571702	0.441334833
36-45	0.49336035	0.440876823

Insight:-

The variation across age groups is relatively small, suggesting that age alone is not a major determinant of customer value or loyalty in this customer base.

Recommendation:-

Retention and growth strategies should primarily focus on customer behavior rather than demographic characteristics.

Analysis 8: Ideal Customer Profile

Q. What does the ideal customer look like?

SQL Query:-

```
SELECT
age_group,
gender,
category,
COUNT(*) AS customers,
AVG(loyalty_score) AS avg_loyalty
FROM customer_intelligence_dataset
WHERE loyalty_segment = 'Champion'
GROUP BY age_group, gender, category
ORDER BY customers DESC;
```

OUTPUT:-

age_group	gender	category	customers	avg_loyalty
55+	Male	Clothing	71	0.692695
55+	Male	Accessories	63	0.683125
46-55	Male	Clothing	53	0.691429
26-35	Male	Clothing	44	0.664943
36-45	Male	Clothing	41	0.684503
18-25	Male	Clothing	41	0.689223
46-55	Male	Accessories	40	0.692471
26-35	Male	Accessories	37	0.657181
36-45	Male	Accessories	34	0.671949
18-25	Male	Accessories	30	0.685318

Insight:-

The Champion customer segment is composed of male customers purchasing Clothing and Accessories. The highest concentration of loyal customers is found among males aged 55+ purchasing Clothing and Accessories. Clothing consistently appears across all age groups, indicating that it is the strongest category for attracting and retaining highly loyal customers.

Recommendation:-

The company should design targeted acquisition and retention campaigns around the characteristics of existing Champion customers. Personalized product recommendations, loyalty rewards, and category-focused marketing campaigns centered on Clothing and Accessories can help attract more customers with similar high-value behaviors.